

Lesson Objective

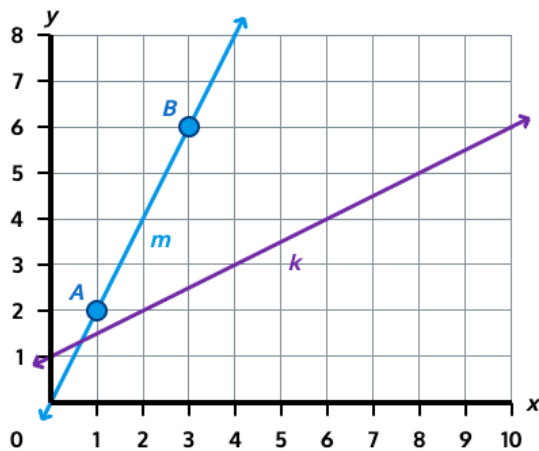
Find the slope of a line on a grid using properties of slope triangles.

Materials: Straightedge

1. Compare the steepness of two lines on a grid and find the slope of one line using a slope triangle.

Which line is steeper, line m or line k ?

Then find the slope of line m . Use points A and B on line m to help.



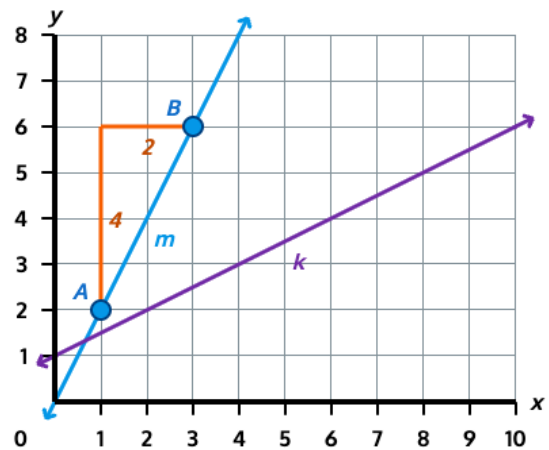
Answer: Line m is steeper; the slope of line m is 2.

Show students lines m and k . Ask which line they would slide down faster, and tell students that line is steeper.

Tell students that the slope of a line is a number that describes its steepness. We can find it by drawing a slope triangle between two points on the line and dividing the vertical distance by the horizontal distance.

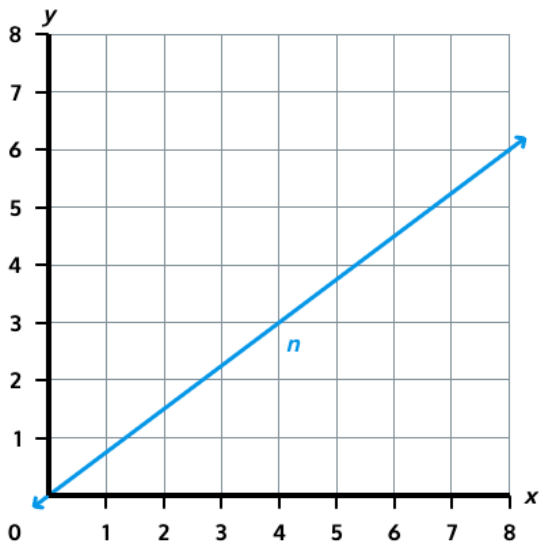
Model drawing a slope triangle from point A to point B with a vertical side of 4 units and a horizontal side of 2 units. Ask students to compute 4 divided by 2 to find the slope.

Point out that the slope of line m is 2, and confirm this matches what they saw: line m is the steeper of the two lines, and it has the larger slope.



2. Find the slope of a line on a grid by drawing a slope triangle between two points.

What is the slope of line n ? Choose two points on the line where grid lines intersect. Draw a slope triangle between them, then find the slope.



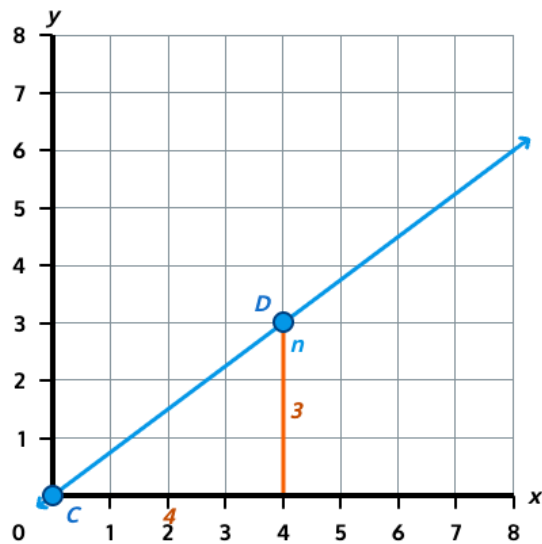
Ask students why it helps to choose points where the grid lines cross. Guide them to see that lattice points let us count the vertical and horizontal sides exactly.

Have students mark two lattice points on line n and draw the slope triangle between them with a straightedge. Ask students to count the vertical side and the horizontal side.

Have students record the vertical distance over the horizontal distance and simplify if needed.

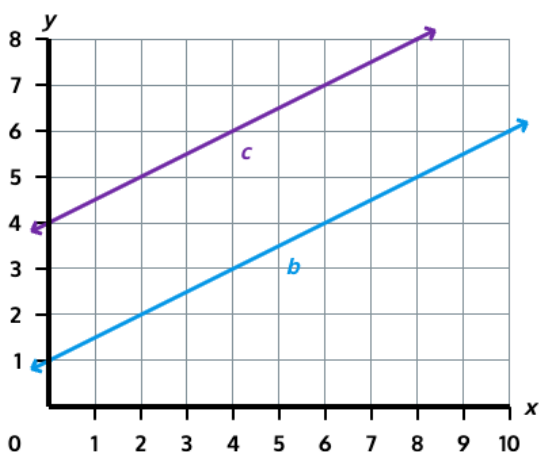
Point out that any two lattice points students choose on line n will give the same slope, because all slope triangles drawn on the same line are similar.

Answer: The slope of line n is $3/4$. (Any two lattice points on line n produce the same slope because all slope triangles on a line are similar.)



3. Find the slope of a line parallel to a line with a known slope.

Lines b and c are parallel. The slope of line b is $1/2$. What is the slope of line c ?



Remind students that lines with the same slope are parallel. Two lines with the same steepness will never meet.

Students work independently to state the slope of line c .

If a student is unsure, prompt them to draw a slope triangle on line c and confirm that the vertical over horizontal distance matches the slope of line b .

Answer: The slope of line c is $\frac{1}{2}$.

Monitor For

- In Problem 1, do students place the vertical distance over the horizontal distance (not the other way around) when writing the slope?
- In Problem 2, do students choose two points that fall on grid line intersections so they can count whole units for the vertical and horizontal sides?
- Do students correctly identify the vertical side and the horizontal side of the slope triangle, rather than the slanted side along the line itself?
- In Problem 3, do students apply the property that parallel lines share the same slope, rather than redrawing a slope triangle from scratch and getting a different number?

Instructional Moves

- Before computing in Problem 1, reinforce that slope is vertical distance divided by horizontal distance by writing the ratio as the fraction bar with "vertical" above and "horizontal" below before any numbers are plugged in.
- If a student chooses two points that do not land on grid intersections in Problem 2, prompt them with, "Can you find two points where the line crosses where the grid lines meet?" so the side lengths are countable.
- Emphasize the conceptual link in Problem 1: a larger slope means a steeper line. State that line m has slope 2 and line k has a smaller slope, which matches what students saw with the slide.
- If a student doubts the slope is the same anywhere on the line, have them draw a second slope triangle between two different lattice points on line n and confirm they get the same quotient.

Reflection Questions

- How did you know line m was steeper than line k before you calculated any slope? *Answer: If I imagined sliding down line m, I would go faster, so it leans more sharply from left to right.*
- Why did you put the vertical distance over the horizontal distance when finding the slope of line n? *Answer: That is how slope is defined. Slope is the vertical change for a given horizontal change between two points on the line.*
- If you had chosen different points on line n, would you have found the same slope? Why? *Answer: Yes. Any slope triangle drawn on the same line is similar to the others, so the vertical over horizontal will always be the same quotient.*
- How did knowing the slope of line b help you find the slope of line c without counting? *Answer: Lines b and c are parallel, so they have the same steepness, which means the same slope.*